

Zehranaz Canfes

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Education

PhD Student | Computer Vision | July 2026 - Present

Computer Vision Group, *Technical University of Munich, Munich, Germany*

Master of Science | Computer Science | October 2022 - March 2026

Technical University of Munich, Munich, Germany

Bachelor of Science (Double Major) | Computer Engineering | September 2018 - June 2022

Bogazici University, Istanbul, Turkey

GPA: 3.45/4.00

Bachelor of Science (Double Major) | Mathematics | September 2017 - June 2022

Bogazici University, Istanbul, Turkey

GPA: 3.45/4.00

Publications

TwoSquared: 4D Generation from 2D Image Pairs, arXiv preprint arXiv:2504.12825, 2025,
Lu Sang, **Zehranaz Canfes**, Dongliang Cao, Riccardo Marin, Florian Bernard, Daniel Cremers
Access paper [here](#).

4Deform: Neural Surface Deformation for Robust Shape Interpolation, **Conference on Computer Vision and Pattern Recognition (CVPR) 2025**,
Lu Sang, **Zehranaz Canfes**, Dongliang Cao, Riccardo Marin, Florian Bernard, Daniel Cremers
Access paper [here](#).

Implicit Neural Surface Deformation with Explicit Velocity Fields, **International Conference on Learning Representations (ICLR) 2025**,
Lu Sang, **Zehranaz Canfes**, Dongliang Cao, Florian Bernard, Daniel Cremers
Access paper [here](#).

Text and Image Guided 3D Avatar Generation and Manipulation, *Spotlight on IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2023*,
Zehranaz Canfes, M. Furkan Atasoy, Alara Dirik, Pinar Yanardag
Access paper [here](#).

Experience

PhD Candidate | 01.07.2026 - Present

Computer Vision Group, *Technical University of Munich, Munich, Germany* | cvg.cit.tum.de

- Research on **4D reconstruction and generation** from 2D keyframes.

Computer Vision Student Researcher | October 2023 - March 2026

Computer Vision Group, *Technical University of Munich, Munich, Germany* | cvg.cit.tum.de

- Working on a **research project** on deformable **3D/4D generation** using **diffusion** models as Master's thesis.
- Completed Guided Research and Interdisciplinary Project (IDP) courses at the Technical University of Munich.

Generative AI Researcher | Internship | April 2024 - October 2024

The BMW Group, *Munich, Germany* | bmwgroup.jobs

- Trained** and **tested** state-of-the-art 3D generative models using different surface representation methods such as **B-reps**, **NURBS**, **point clouds** by using **Python**, **PyTorch**, **PythonOCC**, **occwl**, and **geomdl**.
- Performed **latent space analysis**, advanced **quantitative and qualitative analysis** on 3D generative models using **CATIA**, **scikit-learn**, and **PyTorch3D**, leading to better understanding of the proposed method's behavior. The results are used for further research in BMW.

Undergraduate Researcher | October 2021 - June 2022

Creative AI Technologies Research Lab, *Istanbul, Turkey* | catlab-team.github.io

- Published a paper** on 3D avatar editing guided by text or images by manipulating the latent space of a 3D generative network at the **WACV 2023 conference**. The model is implemented using **Python**, **Tensorflow**, and **PyTorch**, and achieves **34% higher scores** than previous approaches.

Artificial Intelligence Researcher | Internship | July 2021 - September 2021

Università di Bologna, *Bologna, Italy* | corsi.unibo.it

- Proposed a neural network architecture** (autoencoder model) to detect anomalies in a semi-supervised way by using **Python** and **Tensorflow**.
[The proposed architecture](#) increased the **F2-score** by **30%**.

Master's Projects

Computer Vision Practical Course

- Participated in the [practical course: Shape Reconstruction and Matching in Computer Vision](#) at the TUM Computer Vision Group. **Improved** an existing approach to work on **multi-view 3D reconstruction** of objects with non-trivial backgrounds by using **Python**, **Pytorch**, and **Pytorch3D**. The project aimed to be **used by the Computer Vision Group** for further research.

Machine Learning for 3D Geometry

- Adapted the codebase of the paper [3D-LMNet](#), which originally used **TensorFlow 1.3**, to **PyTorch**. The project involved ensuring that the code maintained its original functionality and performance for the task of **single-view reconstruction** of 3D point clouds. The code can be found [here](#).

Certificates and Awards

Scholarship | DAAD-TEV

DAAD-TEV-Master's Degree Scholarship

German Language Certificate | Sprachdiplom Kultusministerkonferenz

Level II, C1